

ROHAN W

Senior Machine Learning Engineer | ML Systems, GPU Kernels, Inference Optimization, Performance Engineering
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Summary

Senior Machine Learning Engineer with 7+ years of experience building and optimizing ML systems across model architecture, GPU kernels, inference runtimes, and production serving. Hands-on experience with CUDA/C++, Triton, PyTorch, TensorRT, ONNX, transformer internals, performance profiling, memory-aware optimization, and large-scale LLM serving. Built custom GPU kernels, implemented GPT-style transformer components from first principles, and designed production inference systems serving 1M+ daily requests.

Technical Skills

- Programming & Kernel Development:** Python, C++, CUDA C++, Triton, PyTorch; custom GPU kernels, parallel execution, tensor indexing, reductions, memory coalescing, data-layout optimization
- ML Systems & Inference:** vLLM, SGLang, TensorRT, ONNX Runtime, CUDA Graphs, FP8/FP16 quantization, dynamic batching, KV-cache sizing, scheduling, latency/throughput optimization
- Transformer & ML Architecture:** causal self-attention, multi-head attention, LayerNorm, GELU, feed-forward networks, residual connections, LoRA, BERT, GPT-style decoder models
- Performance & Infrastructure:** profiling, benchmarking, bottleneck analysis, GPU utilization, memory-footprint analysis, Docker, Kubernetes, Helm, CI/CD, distributed workflows, AWS/Azure/GCP

Experience

Senior Machine Learning Engineer

March 2025 – Present

Avalara Bengaluru, IN (Remote)

- Designed production-grade **GLM-5.2 FP8 inference serving** on **2× AWS p5e.48xlarge**, analyzing model/KV-cache memory footprint, batching and concurrency limits, routing, failure isolation, and latency/throughput trade-offs.
- Built and deployed a multi-cloud LLM serving platform exposing **OpenAI-compatible APIs** across Azure, AWS Bedrock, and Google Vertex AI, handling **1M+ daily requests** for internal GenAI workloads.
- Built a **BERT-based multi-label classification** system with label design, threshold tuning, class-imbalance handling, controlled evaluation, and production validation.
- Profiled and optimized GPU inference paths using **TensorRT, ONNX, batching, and runtime scheduling** to improve latency, throughput, and accelerator utilization for classification workloads.

ML Research Engineer - II (L5)

April 2024 – March 2025

Amazon Bengaluru, IN

- Built a production pipeline fine-tuning **Mistral 7B with LoRA and Hugging Face** and serving it through **vLLM** for catalog enrichment, with model-quality evaluation against a Claude-based baseline.
- Improved Items Glance View by **20% across 100M products** through controlled experimentation, model-assisted defect detection, A/B testing, and production deployment.
- Developed an LLM platform using **Claude 3.5 Haiku** for missing-attribute generation and catalog backfilling at production scale.
- Built an automated prompt-optimization POC using **DSPy**, projected to improve ML engineer productivity by **140%**; recognized with **RISE Q4 2024**.

Senior Machine Learning Engineer

Feb 2020 – March 2024

Rapid Acceleration Partners Chennai, IN

- Wrote custom **CUDA C++ kernels** for PaddleOCR preprocessing and post-processing, including GPU-side normalization, **HWC→CHW data-layout transformation**, and DBNet probability-map binarization; integrated the optimized path with **TensorRT FP16**.
- Re-architected ML inference pipelines for GPU execution, optimizing runtime/export paths, batching, data movement, and model execution for production document-AI workloads.
- Built GPU-optimized document classification and understanding pipelines using **TensorRT, ONNX, and batching**, focusing on latency, throughput, and efficient accelerator execution.
- Led and mentored a **4-member ML team** delivering document AI, OCR, classification, and automation systems for production customers.

Data Science Associate

Dec 2018 – Nov 2019

ZoomRx Chennai, IN

- Built Python data-processing and analytics tooling integrating multiple upstream data sources into reliable downstream workflows.

Selected Systems Projects

LLM From Scratch | PyTorch, Transformer Internals, LoRA

- Implemented a GPT-style decoder model from first principles in PyTorch, including **causal multi-head attention, Q/K/V projections, masking, LayerNorm, GELU feed-forward blocks, residual connections, embeddings, and autoregressive generation**; trained and evaluated the model and implemented LoRA components.

GPU Kernel Engineering | Triton, CUDA, PyTorch

- Implemented and validated custom **Triton kernels** for vector operations, reductions/row-wise operations, transpose and tiled tensor workloads; analyzed strides, masking, launch configuration, memory access patterns, and coalescing against PyTorch reference implementations.

Education

SRM University

July 2015 – July 2019

B.Tech, Electronics and Communication Engineering

Chennai, TN

Open Source

- Contributor to open-source ML and systems libraries, with merged pull requests across Python runtime, LLM infrastructure tooling, and optimization frameworks. Contribution history available on GitHub.